SELF-STUDY PROJECT REPORT • MINI NLP ASSIGNMENT

Automated Spam Detection System Using TF-IDF Vectorization & Machine Learning

DOMAIN	DATASET	PRIMARY MODEL	IMPLEMENTATION
Natural Language Processing	**UCI SMS Spam Collection**	**Multinomial Naive Bayes**	**Python / Scikit-Learn**

1. EXECUTIVE SUMMARY & TITLE

This project presents an automated Natural Language Processing (NLP) framework designed to detect and filter unsolicited electronic spam messages. By leveraging custom text normalization, Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction, and supervised machine learning algorithms, the system automatically categorizes text inputs as either legitimate ("Ham") or "Spam" with high operational precision.

2. PROBLEM STATEMENT

With the exponential growth of electronic communications via SMS and digital messaging, individuals and organizations face increasing risks from fraudulent messaging, phishing scams, and mass unsolicited promotional spam. Manual content moderation is impossible at scale. Therefore, developing an automated machine-learning pipeline capable of contextualizing and categorizing unstructured text messages in real-time is an essential practical NLP task.

3. OBJECTIVES

- **Text Normalization:** Preprocess raw, unstructured text messages through cleaning, tokenization, stop-word removal, and stemming.
- **Feature Extraction:** Convert normalized text sequences into sparse numerical vector representations using TF-IDF.
- **Model Training & Benchmarking:** Implement and compare three classifiers: Multinomial Naive Bayes, Logistic Regression, and Support Vector Machines (SVM).
- **Performance Evaluation:** Assess classification performance using Accuracy, Precision, Recall, and F1-Score metrics to ensure minimal false positives.

4. DATASET SPECIFICATIONS

The project utilizes the public **UCI SMS Spam Collection Dataset**, recognized as a benchmark resource for text classification tasks.

Attribute	Specification Details
Dataset Source	UCI Machine Learning Repository
Total Sample Count	5,574 text messages (100%)
Target Classes	Binary: Legitimate (`ham`) vs. Unsolicited (`spam`)
Class Distribution	Ham: 4,827 samples (86.6%) | Spam: 747 samples (13.4%)
Data Partitioning	80% Training Set (4,459 samples) | 20% Testing Set (1,115 samples)

5. PIPELINE & METHODOLOGY

6. IMPLEMENTATION DETAILS

The complete software system was built entirely in **Python 3** using core scientific computing and NLP libraries:

- **Data Handling & Analysis:** `pandas` and `numpy` for dataset manipulation and matrix operations.
- **NLP Preprocessing:** `nltk` for stop-word resources and morphological stemming algorithms.
- **Machine Learning:** `scikit-learn` for TF-IDF feature extraction (top 3,000 features) and model architectures.
- **Data Visualization:** `matplotlib` and `seaborn` for visual confusion matrix generation.

7. MODEL PERFORMANCE & EVALUATION RESULTS

All three supervised machine learning algorithms were trained on the same 80% split and evaluated on the 20% holdout test dataset (1,115 samples):

Algorithm / Model	Accuracy	Precision	Recall	F1-Score
Multinomial Naive Bayes	**97.13%**	**98.44%**	80.00%	88.25%
Logistic Regression	96.14%	97.35%	73.33%	83.65%
Support Vector Machine (Linear)	97.40%	96.88%	82.67%	**89.21%**

97.13%
Top Overall Accuracy

98.44%
Naive Bayes Precision

89.21%
Peak SVM F1-Score

Key Finding: Multinomial Naive Bayes achieved an extraordinary **98.44% Precision**, ensuring that legitimate user messages ("ham") are almost never wrongly flagged as spam—a crucial requirement for practical real-world deployment.

8. LIMITATIONS

- **Domain Restriction:** The model was trained on short SMS data and may perform less effectively on long, structured multi-paragraph email threads.
- **Adversarial Evasion:** Spammers employing intentional character misspellings or zero-width spaces may evade unigram word stemmers.
- **Fixed Vocabulary:** Feature extraction is restricted to the top 3,000 TF-IDF features, omitting rare context words.

9. FUTURE SCOPE

- **Contextual Transformer Embeddings:** Transitioning from TF-IDF feature matrices to pre-trained transformer architectures (e.g., BERT, DistilBERT).
- **Web Deployment:** Wrapping the inference model in a lightweight FastAPI microservice integrated with a Streamlit interface.
- **Multilingual Support:** Expanding dataset training to detect cross-lingual and translated spam sequences.

10. CONCLUSION

This self-study project successfully implemented and evaluated an end-to-end NLP classification pipeline for spam detection. By pairing standard text normalization techniques with TF-IDF vectorization and Multinomial Naive Bayes / SVM classifiers, the system achieved over 97% classification accuracy. The project demonstrates how fundamental classical NLP approaches provide lightweight, robust, and highly accurate solutions for everyday text classification problems.